Language side and the integrator framing: expansion for Section 7 and the Discussion (2026-10-20)

Scope: what a one-layer path model learns from words, when a learned phase becomes a map and when a clock, the integrator (Howard) framing, and whether what/where and the leak belong in the main paper. For `docs/paper/MAIN_DRAFT1.md` Sections 2.1, 7 and 8. No training was run; every new number is a post hoc, eval-only CPU analysis of stored checkpoints (scripts and outputs in `docs/paper/expansion/scripts/language/`, section 8). Results files are authoritative; where this note and a results file differ, the results file wins.

Labels. **ESTABLISHED** = a registered verdict or a declared secondary in a results file (with its file). **ANALYSIS** = post hoc, on stored checkpoints, not pre-registered (new here unless a file is named). **THEORY** = an argument, with the check that tests it. **SPECULATION** = untested. Terms: *step* = the per-token increment Delta_w = W_out W_in e_w of a word w (one per word type; the model's step is context-free); *move* = one step of the walk (one movement clause in the text world); *tick* tau_c = the phase a channel gains per move whatever its direction; *channel* = one rotary 2D block of one head (64 per model); *used channel* = weighted by how much retrieval attention uses it (mean ||q_c|| x mean ||k_c|| over the query words and the object words); *re-scored* = read with the x1.111 attention-dropout scale (the registered correction, `docs/audits/2026-10-04/DROPOUT_RESCORE.md`).

---

0. Summary: the key additions

1. **A complete step account of the text-world models (ANALYSIS, 32 models).** One path layer learns: direction words as signed steps whose differentials cancel (opposition 0.001-0.013 on all 24 learned-step models, wrap-aware); optional words (adverbs, fillers) with steps of 0.004-0.03 of a move on every model; aside sentences whose words have sizeable steps that cancel within the sentence, leaving a net offset of 0.04-0.16 of a move (usage-weighted); and, on 10 of 24 models, a per-move tick in 26-38 channels.
2. **The aside offset is causal, both ways (ANALYSIS).** Putting each aside's words at its clause's object-slot phase, with the phase after the aside unchanged, drops the 19 learned-step models at ceiling in eval mode from 0.989-1.000 to 0.952-0.975, and 71-100% of their errors then name an object told in an aside at that cell: the direction-words-only oracle's ceiling and error signature (`TW_NORMSTEP_RESULTS.md`). Conversely, offsetting the oracle's aside words by half a move (phase after the aside unchanged) lifts it from 0.962-0.981 (100% aside errors, 8/8 seeds) to 0.999-1.000 on 8/8 seeds.
3. **The text-world clock counts moves, so it is path length, Howard's Case II (ANALYSIS).** An aside sentence adds at most 0.06 of a tick, a filler or adverb word at most 0.007: the clock is neither a word nor a sentence counter. On the clock seeds one phase runs Case III (position) in some channels and Case II (path length) in others.

4. **The clock is idle (ANALYSIS).** The per-move tick alone predicts which channels drift between visits (agreement 0.98-1.00 per model, 32 models). 79-99% of retrieval usage sits in channels whose tick is 0 mod pi (pi: on the even 64-torus every return takes an even number of moves). Removing the 26-38 clock channels from the score costs <= 0.009 on all 7 clock models at ceiling in eval mode, while the 3 below ceiling still need them (cost 0.09-0.26); removing as many map channels collapses accuracy (0.02-0.65).
5. **A usage-weighted drift/content ratio separates map from clock where the growth exponent cannot (THEORY + ANALYSIS).** The pooled growth exponent alpha reads 0.98-1.02 on the text-world clock seeds (which carry a perfect map) and on TinyStories alike. The usage-weighted median of rho_c = |mu_c| L / range(content_c, L) (clock part over content part, per channel, at the training length) reads: torus 0.064-0.079, text world 0.06-0.23 (converged seeds), code 4.5-6.6, TinyStories 9.9-12.3, recency 21.8-22.1. Same architecture, same signed step: the data decide whether the used channels are a map or a clock.
6. **The text world is a short-walk task, and the paper must say so (ANALYSIS).** Its walks are ~132 moves, and the paper's own retrace floor scores **0.934** on its registered eval stream (0.937 on TW_NORMSTEP's), not the 0.597 of the one-step reversal-copy rule the results file reports. Path still clears it, and on the revisits retrace gets wrong (5.9%; best constant 0.112) path models score 0.944-1.000 on the six seeds at ceiling (0.961 on the fresh seeds 2-7) against 0.10-0.17 for two index layers and 0.00-0.01 for one (section 2.6).
7. **Integrator paragraph** (section 4): a positive statement of what is ours (the three-way identification for rotary attention; that in the rotary slot the Case II / III distinction is one of learnability, not expressivity; and the measurement that a learned rate realises Case I, II or III per channel as the data demand), with precise prior work.
8. **What/where and the leak** (section 5): one paragraph in Section 7, as a single principle -- a token's phase offset from its place must depend on its role, not its identity -- with the leak registered result (and its failed accuracy replication) and the what/where checks in the appendix.

---

1. Established facts (from the results files)

fact	numbers	label	file
One path layer learns navigation told in English	fresh seeds (n = 6): 0.979 +/- 0.048, 6/6 SOLVED vs RoPE 1 layer 0.505, 0/6; +0.474, permutation and Fisher p 0.0022; RoPE 2 layers 0.793 (0/6). All 8 seeds: 0.969 (eval mode), **0.993 re-scored** vs 0.506 / 0.772	ESTABLISHED (A; the re-score is registered in DROPOUT_RESCORE)	`TEXTWORLD_RESULTS.md`
Floors on that eval stream	best constant 0.505, one-step reversal-copy 0.597	ESTABLISHED	`TEXTWORLD_SECONDARY.txt`
Opposite directions	after removing the direction words' common	ESTABLISHED (declared secondaries; registered	`TEXTWORLD_RESULTS.md`

fact	numbers	label	file
cancel on every seed	component: 0.002-0.012 (8/8); synonyms one step (cosine 1.000, vs 0.36-0.61 across directions); zeroing direction steps -> 0.39-0.51	B branch did not fire, 4/8)	
Half the seeds also count	4/8 seeds: common component parallel to the verbs' step (cos +0.999 to +1.000); 31-38 of 64 channels drift between visits (2-4 in map seeds)	ESTABLISHED (secondary + CORRECTED block)	`TEXTWORLD_RESULTS.md`
Learned steps beat "only direction words move"	DirOnly 0.970-0.974 on 8/8 seeds; on 2 seeds 100% of its errors name an aside object told at that cell; learned arms make none. Re-scored DirOnly - MapWM -0.024 (p 0.0002), a post hoc correction (eval mode: no difference)	ESTABLISHED (post hoc in the file)	`TW_NORMSTEP_RESULTS.md`
NormStep on words	accuracy +0.006, UNMEASURED below 0.096; the registered "per-word clock" is the aside offset (0.126 of 0.133 rad)	ESTABLISHED	`TW_NORMSTEP_RESULTS.md`
State sentences	MapWM 0.994 vs RoPE 0.516 (sanity check); C = location + recency among nouns at a place, NOT verb semantics (a verb-agnostic rule scores 1.000 on T2drop); state sentences displace the phase 0.028 moves, like asides; DirOnly worse only under the re-score	ESTABLISHED (CORRECTED 2026-10-11)	`TW_STATECHANGE_RESULTS.md`
What/where, causal	converged path models keep 0.974-0.989 with object identity removed from the score, 0.988-1.000 with a shared kernel x content gain; index models fall below the floor	ANALYSIS (post hoc)	`docs/WHAT_WHERE_CHECKS.md`
The what-to-where leak	in distribution, removing it (ActOnly / NormStep) gains +0.0107 (p 0.0002, 8/8), equal to MapWM's leak; both remedies SOLVED 16/16 vs MapWM descending. Fresh seeds: SOLVED replicates (8/8 vs 0/8), accuracy does not (+0.0045, p 0.46)	ESTABLISHED (LEAK registered; GAIN_PHASE D1r)	`LEAK_RESULTS.md`, `GAIN_PHASE_RESULTS.md`

fact	numbers	label	file
TinyStories	MapWM 1.5053 vs RoPE 1.5050 nats/token (n = 3); clock share 0.79-0.87; `mr`/`mrs` cancel '.'; a uniform clock costs 0.07-0.09 nats	PILOT (CORRECTED 2026-10-11)	`TINYSTORIES_PILOT_RESULTS.md`
Growth exponent on text and code	alpha 0.999-1.006 (TinyStories), 0.990-1.016 (code); R^2(S ~ t) 0.994-0.999	ANALYSIS	`docs/audits/2026-10-12/alpha_language_out.txt`
Code at matched length	position main +0.0056 bpc (3/3, t p 0.024): path integration costs on code	ESTABLISHED (n = 3)	`CODE_FULLVAL_RESULTS.md`

Not to be revived (CLAUDE.md, withdrawn): "the model tracks state changes"; TinyStories "nothing cancels", "spatial 0%", "20x the MapWM - RoPE difference"; "two defects, two fixes".

---

2. What a one-layer path model learns from words (question 1)

Models: the 8 registered text-world path runs (`runs/textworld/p0`, seeds 0-7) and the 8 MapWM, 8 NormStep and 8 DirOnly runs of TW_NORMSTEP (`runs/tw_normstep/p0`, seeds 10-17). Eval: held-out map, np seed 424242 (used by no registered script), 40 walks x 1024 words, unless stated. The manual forward used for manipulations reproduces every model's logits exactly (max |diff| 0.0) and the role-annotated generator reproduces `TextWorld.generate_trajectory` token for token (`tw_lib.py` checks).

2.1 The step table, by word class (ANALYSIS; `tw_step_account_out.txt`)

Steps omega-scaled (radians), relative to the clause-level direction differential |D_N| (one move):

word class	map-type models (14 learned-step)	clock-type models (10)	what it means
direction words (differential)	1.00 by normalisation; opposites cancel 0.001-0.013	same	signed, cancelling steps
adverbs, fillers	0.004-0.03	0.004-0.025	words whose count per move varies get no step
`and` / `a` / `she` (shared by fillers, seeing phrases, asides)	0.02-0.07	0.02-0.08	near zero
verbs, seeing words, object words, `.`	0.02-0.14	0.24-1.49	one per move: free to carry a per-move tick
aside-only words (`thought`, `about`, `remembered`, `story`)	0.07-0.16 per word	0.41-1.33 per word	large, but cancel within the sentence
aside sentence, net (usage-weighted)	0.05-0.12 of a move	0.04-0.16 of a move	a small offset off the place
DirOnly (oracle): every non-direction word	0 by construction	--	asides sit at the place's phase

The clause tick tau (what every move adds whatever its direction) is 0.03-0.23 of a move (map type) and 3.3-6.8 moves (clock type, raw norm, all channels), parallel to the verbs' step (cos +0.999 to +1.000 on all 10 clock models).

2.2 The tick predicts the drift; the clock counts moves (ANALYSIS)

- **The tick alone predicts which channels drift.** Predicted drift per channel = mean |wrap(lag x tau_c)| over the consecutive-visit pairs (median lag 8 moves, 1,221 pairs); observed = mean |wrap(dtheta_c)| at those pairs. Classified at 1 rad, they agree on 0.98-1.00 of channels on every one of the 32 models (r 0.96-1.00 on the 24 learned-step models). The clock seeds drift in 26-38 channels (tw s0/s2/s3/s4: 26/29/35/36; MapWM s11/13/14/15: 38/27/28/30; NormStep s11/16: 27/28), map seeds in 0-3. (The CORRECTED block's 31-38 counted drift from the FIRST visit; consecutive visits give the numbers here.) `tw_step_account_out.txt`.
- **It counts moves, not words or sentences.** Regressing each move's net phase on direction, aside present, filler count and adverb present, in the drifting channels: an aside sentence adds 0.009-0.062 of a tick, a filler word 0.001-0.007, an adverb word 0.001-0.007 (10 clock models; `tw_clock_unit_out.txt`). A word counter would give fillers and adverbs ~1/8 of a tick each; a sentence counter would give asides ~1 tick. Every move is one cell, so the move count is the path length: **these channels integrate Howard's Case II (distance travelled) while the others integrate Case III (position)**.
- **Pi ticks are map features.** On all 10 clock models 2-6 channels tick by exactly pi per move (e.g. MapWM s11: -3.141, -3.140 rad). On the even 64-torus every return to a cell takes an even number of moves, so a pi tick is invisible to revisits: it encodes the checkerboard parity of the cell. These channels carry up to 0.24 of retrieval usage (MapWM s11).
- **Why clock seeds lean on non-direction words.** The registered secondary found that zeroing every non-direction step costs the clock seeds 0.33-0.69 and the map seeds 0.05-0.30. In the channels retrieval uses (tick 0 mod pi), the direction words' common component is 0.06-0.34 of a move on the clock seeds and 0.01-0.03 on the map seeds (registered text-world batch), and the other words cancel it within the move. Zeroing them leaves it as a tick in exactly the used channels: across the 8 seeds the cost tracks that component (Spearman +0.905; `tw_common_used_out.txt`).

2.3 The clock is idle: usage and channel ablation (ANALYSIS)

- **Usage.** 79-99% of retrieval usage sits in channels whose tick is 0 mod pi (lowest: tw s0 0.79, unconverged in eval mode; NormStep s16 0.82); channels where the clock dominates the phase (rho_c > 1 and more than half a turn per window, section 3) carry 0.13-0.30 of usage on clock models and 0.01-0.15 on map models (`usage_tick_out.txt`).
- **Ablation.** Removing the drifting channels' q and k components from the score (re-scored accuracy):

model	intact	clock channels removed	same number of map channels removed (5 draws)	only clock channels kept
	0.973	**0.717**	0.43 [0.24, 0.58]	0.07

model	intact	clock channels removed	same number of map channels removed (5 draws)	only clock channels kept
tw s0 (eval mode 0.875, unconverged)				
tw s2	0.999	0.990	0.25 [0.03, 0.63]	0.03
tw s3	0.998	0.993	0.06 (all 29 map channels)	0.06
tw s4	0.999	0.999	0.03 (all 28)	0.03
MapWM s11	0.989	0.985	0.48 (all 26)	0.48
MapWM s13	0.997	0.996	0.65 [0.50, 0.84]	0.48
MapWM s14 (eval mode 0.960)	0.995	**0.902**	0.25 [0.03, 0.72]	0.03
MapWM s15	0.995	0.994	0.07 [0.02, 0.23]	0.02
NormStep s11	0.998	0.996	0.46 [0.20, 0.60]	0.44
NormStep s16 (eval mode 0.835)	0.982	**0.809**	0.22 [0.04, 0.48]	0.04

On the 7 clock models at ceiling in eval mode the clock costs at most 0.009 to remove; the 3 that need it (tw s0, MapWM s14, NormStep s16: 0.09-0.26) are exactly the ones below ceiling in eval mode, i.e. not converged to the clean solution. A clock in the channels retrieval does not use is how half the seeds and a perfect map coexist. That unconverged models still lean on it suggests the clock is used early and abandoned later (SPECULATION; E1 tests it). Why training builds it at all is not identified.

2.4 Asides: the offset is what beats the oracle (ANALYSIS; `tw_attention_asides_out.txt`)

`TW_NORMSTEP_RESULTS.md` found that DirOnly stops at 0.970-0.974 because an aside's noun sits at the place's phase, and that learned-step models make none of these errors. Two manipulations at eval, each keeping the phase after the aside unchanged (so only the within-sentence offset changes):

- **Collapse** (learned models): every aside word is put at its clause's object-slot phase. The 19 learned-step models at ceiling in eval mode (>= 0.98) fall from 0.989-1.000 to **0.952-0.975**, and **71-100%** of their errors then name an object told in an aside at that cell (tw s1-s6 0.94-1.00; MapWM s10, s11, s13, s15-s17 0.71-1.00; NormStep s10-s15, s17 0.86-1.00); before the manipulation they make 0-13 errors, none of that kind. That is DirOnly's ceiling and its error signature, produced in the models that had escaped it. (The 5 models below ceiling in eval mode respond partly: 0.966-0.994, 0-85% aside errors.)
- **Lift** (DirOnly, `tw_attention_asides_dironly_out.txt`): intact, DirOnly scores 0.962-0.981 and **100%** of its errors are aside errors on **8/8** seeds (TW_NORMSTEP checked 2). Offsetting its aside words by half a move in-plane, with the phase after the aside unchanged, gives **0.999-1.000 on 8/8**; by one move off the map plane (a random direction orthogonal to the direction steps) 1.000 on 7/8 (s16 0.972); by a quarter move off-plane 0.963-0.996 (the kernel is not sharp at that distance). The oracle's ceiling is exactly the missing aside offset.

So the learned non-direction steps do two jobs: they cancel within every move (so the place's phase is the same whatever the wording) and they leave a small net offset inside aside sentences (so a mentioned-but-absent object is not filed at the place).

2.5 Attention at revisits (ANALYSIS)

At each revisit query (head mean), the 19 learned-step models at ceiling put 0.55-0.76 of their attention on the object slot of the most recent visit to that cell, 0.07-0.14 on older visits' object slots, 0.004-0.035 (mean 0.015) on aside nouns told at the cell, 0.07-0.30 on other tokens of earlier clauses at the cell, and 0.01-0.06 elsewhere. Retrieval is by place: 0.94-0.99 of the attention lands on tokens told at the current cell. DirOnly puts 0.041-0.077 (mean 0.047, about 3x) on aside nouns told at the cell and 0.10-0.23 elsewhere.

2.6 The text world is a short-walk task: the retrace floor (ANALYSIS; `tw_retrace_floor_out.txt`, `tw_strata_out.txt`)

The text world renders ~132 moves per 1024 words, the walk length at which the main paper's Section 4 shows a map-free retrace rule scoring 0.934 on the torus. On the text world's own registered eval stream the same rule (port of `docs/audits/2026-10-16/sign_floors.py`) scores **0.9337** (TW_NORMSTEP stream: 0.9365), while the constant and the one-step reversal-copy rule reproduce the registered 0.5054 / 0.5965 exactly. Consequences:

- The text world's accuracy margin over the best map-free rule is small: path 0.993 re-scored (0.969 eval mode) vs retrace 0.934; DirOnly's 0.972 is barely above it. Index RoPE (0.505 / 0.772) does not reach retrace.
- The map claim must rest on the revisits retrace gets wrong and on the mechanism (cancelling steps; direction steps zeroed -> floor). On the first 100 walks of the registered stream, 179 of 3,055 revisit targets (5.9%) are ones retrace gets wrong (best constant there 0.112: 'cat'; 'nothing' never):

model (registered batch, n = 8)	all	RETRACE-OK	NEEDS-MAP (n 179)
path, 1 layer, re-scored	0.994	0.998	**0.920** (seeds 1-6: 0.944-1.000; s0 0.603, s7 0.832, the two below ceiling in eval mode); fresh seeds 2-7: 0.961
path, 1 layer, eval mode	0.970	0.975	0.885
RoPE, 2 layers	0.779	0.819	**0.138** (0.10-0.17)
RoPE, 1 layer	0.507	0.538	0.002

Two-layer index RoPE learns part of retrace (two seeds reach 0.93, just under the 0.934 floor; RETRACE-OK 0.98) and stays at the constant floor where a map is needed. Path solves those revisits. This is the text-world analogue of Section 4's strata and the strongest single piece of evidence that the words are integrated into a map. - Section 7 of the draft should report retrace beside the constant floor. This does not change any registered verdict.

---

3. Theory: when a learned phase is a map and when it is a clock (question 2)

3.1 Setting and decomposition (THEORY)

Channel c's score between query t and key t' depends on position only through omega_c (Phi_c(t) - Phi_c(t')), the summed step over the interval. With token frequencies f_w, split every step into a common part and a content part: Delta_c(w) = mu_c + d_c(w), mu_c = sum_w f_w Delta_c(w), so sum_w f_w d_c(w) = 0. Over an interval of L tokens

```
Phi_c(t) - Phi_c(t') = mu_c L + sum of d_c over the interval.
```

The first term is a clock (Howard Case I if mu is carried by every token; Case II if only by selected tokens, as in the text-world move counter); the second is content-driven and can be a map (Case III) only if it cancels over closed loops. "The common component tracks the frequency-weighted mean step" is true by definition of mu; the substance is in which channels mu_c survives and why.

3.2 T1: the tick rule for a retrieval channel (THEORY, checked)

If the target at t is the key at the last visit to the same latent state z (a cell), matched pairs come at lags spread over a distribution, and for a channel to score matched pairs consistently its phase difference over them must be constant mod 2 pi. With the map part cancelling over the closed loop, the remainder is lag x tau_c. Hence: **a channel can serve latent-keyed retrieval only if its tick is 0 mod 2 pi / g**, g the gcd of the possible return lags (g = 2 on a bipartite lattice such as the even torus), and otherwise it decorrelates once |wrap(tau_c)| x sd(lag) exceeds ~1 rad.

Checks (section 2.2-2.3): the tick alone predicts each channel's revisit drift (agreement 0.98-1.00, 32 models); used channels have tick 0 mod pi (79-99% of usage), including exact-pi parity channels; channels that violate T1 are not used, and removing them costs <= 0.01 on converged models.

3.3 T2: the readout decides, channel by channel (THEORY + ANALYSIS)

T1 constrains mu_c only where the readout looks. Elsewhere a tick is free, and the text world shows one surviving in 26-38 channels on 10 of 24 models without cost. Conversely, when targets are keyed to a count back (k-back recency) or to recent context (language), a nonzero mu_c in the used channels is what the readout needs. So the prediction is about **used** channels: their drift/content ratio rho_c = |mu_c| L / range(content_c, L) at the training length L should be << 1 when targets are keyed to a latent state reached at variable lags, and >> 1 when they are keyed to a token or event count.

Check (`crossover_usage_out.txt`, `usage_tick_out.txt`; usage-weighted median rho over channels, all-token usage for the multi-layer language models):

data (all MapWM, signed step)	used-channel rho (median)	usage in clock-dominated channels	seeds
2D torus, 1 layer, matched length (sign_matched Signed_r4)	0.064-0.079	0.000 (0/64 channels)	8

data (all MapWM, signed step)	used-channel rho (median)	usage in clock-dominated channels	seeds
text world, 1 layer (24 learned-step models; tw s0 unconverged 2.5)	0.06-0.23	0.01-0.30	24
code, bytes, 9 layers (trained at 2048)	4.5-6.6	0.50-0.55	3
TinyStories, words, 6 layers (trained at 512)	9.9-12.3	0.42-0.43	3
k-back recency with fillers, 1 layer (recency Signed_r4)	21.8-22.1	0.93-1.00	8

The same architecture and parameterisation (the torus and recency rows are the identical `Signed_r4` class) put its used channels at opposite ends; language and code sit on the recency side. (Clock-dominated: rho_c > 1 and more than half a turn per training window. On language part of the usage is in slow channels with rho > 1 that turn less than half a turn per window -- slow recency kernels -- which is why the share is 0.42-0.55 while the median rho is 4.5-12.)

3.4 What the growth exponent can and cannot show (ANALYSIS; `tw_alpha_drift_out.txt`)

The pooled exponent alpha (range of the raw summed step ~ T^alpha, the readout of `alpha_language.py`) weights channels by their raw range, so a large tick in channels attention ignores dominates it:

models	alpha	pooled rho(1024)	used-channel rho	accuracy
text-world map seeds (14)	0.61-0.74 (the true walk: 0.668)	0.12-0.86	0.06-0.13	0.985-1.000 (re-scored)
text-world clock seeds (10)	0.98-1.02	9.3-13.9	0.07-0.23 (tw s0, unconverged: 2.5)	0.989-0.999 at ceiling (7)
DirOnly (8)	0.60-0.65	0.05-0.07	--	0.962-0.981 (this stream)
TinyStories (3)	1.00-1.01	11.9-12.5	9.9-12.3	MapWM = RoPE
code (3)	0.99-1.01	2.9-5.0	4.5-6.6	path costs +0.0056 bpc

The clock seeds and TinyStories are indistinguishable by alpha and pooled rho, yet one carries a perfect map. So the draft's "it grows linearly with token index (R^2 0.994-0.999)" does not by itself show the absence of a map. The used-channel rho does separate them (0.07-0.23 vs 9.9-12.3): on TinyStories the channels attention uses are dominated by the clock part. The two-part model accounts for alpha: the exponent of |mu| T + range(content, T) is within 0.01-0.07 of the measured alpha on every model (it over-predicts by 0.01-0.03 on map-type models and under-predicts by 0.03-0.07 on clock-type models, TinyStories and code, since the range of a sum is not the sum of the ranges).

3.5 Howard's cases in one model (THEORY + ANALYSIS)

A signed, content-dependent rate can realise all three of Howard et al.'s cases: Case I (mu carried by every token), Case II (a non-negative tick on selected tokens: the text-world move counter; the TinyStories punctuation and boundary ticks read as an event

counter of the same kind, an interpretation), Case III (signed steps that cancel). Which one a channel ends in is set by what the readout keys on, not by the parameterisation: torus and text world put their used channels in Case III, recency and language in Case I/II. The sign result of Section 6.2 is the complement: a non-negative step can represent Case III in the rotary slot (Proposition 1) but does not learn it from a small start.

SPECULATION, not tested: language would put used channels in Case III wherever prediction pays for a latent state reached by variable routes (dialogue about places, procedural text). The text world says a one-layer model finds it when the words that move are context-free; TW_AMBIG (section 7, E2) would test the context-dependent case.

---

4. The integrator framing (question 3)

4.1 What is prior art (from `docs/lit/LIT_HOWARD_TRANSFORMER.md, LIT_CLOCK_MAP.md`)

- Howard et al. 2014 (J Neurosci 34:4692): one integrator bank driven by a rate alpha(t) codes time (alpha constant, Case I), distance travelled (alpha = speed, Case II) or position (alpha = signed velocity, Case III: "like a bead on a wire"). The dichotomy, with the collision property, is theirs.
- Velocity-modulated integrators in trained networks: Alipour et al. 2025 (J Comput Neurosci 53:131; Cases I and III, alpha given, real s); Mochizuki-Freeman et al. 2023 (IJCNN; alpha = dz/dt from the environment). The Howard and Tiganj labs keep s real and explicitly reject complex s for stability (Mochizuki-Freeman 2023 fn. 1; Kabir et al. 2025, AAAI; Dickson et al. 2025).
- RoPE as a velocity-controlled oscillator: GridPE (Li et al., arXiv:2406.07049): Phi(t) = omega_s t + beta omega^T x(t), reducing to RoPE in 1D; coordinates given, not learned from content.
- Laplace memory and SSMs: Sarkar and Howard 2026 (arXiv:2601.02618) relate SITH's diagonal recurrence to discretised SSMs (decay side, not the step Delta_t, not phase).
- Content-gated phase rate motivated by Howard: RoMem (Li et al. 2026, EMNLP; arXiv:2604.11544), monotone (Case II-like), for temporal knowledge graphs.
- Delta_t as a learned stand-in for time or arc length: TIDES (Soydan et al. 2026, arXiv:2605.09742), no signed case.
- Index position as "linear structure" and path-integrated position as a learned map in a transformer: TEM-t (Whittington et al. 2022). Sign and state tracking: Sarrof et al. 2024, Grazzi et al. 2025 (Thm 3 for rotations), Mamba-3 Sec. 3.2.

4.2 What is ours

1. The explicit identification, for rotary attention: index RoPE, monotone content-dependent phase (CoPE, CARoPE, softplus or |.| steps) and signed content-dependent phase (MapFormer, Selective RoPE, Mamba-3's data-dependent RoPE) are Howard's Cases I, II and III with an imaginary rate, and Howard's alpha(t) is the selective-SSM step Delta_t and MapFormer's content-dependent angle step. Not found stated (LIT_HOWARD_TRANSFORMER sec. 1); one sentence once the pieces are side by side, so present it as an organising remark, not a result.

2. The rotary slot is where the signed rate is bounded (a rotation), the move the Howard lab declined for real s; and in that slot Cases II and III coincide in what they can represent (Proposition 1, extending Grazzi Thm 3 / Mamba-3 to rank-limited step maps). The distinction becomes one of learnability (Section 6.2).
3. Measured: which case a learned signed rate realises is decided per channel by the data (section 3.3), and one model can run Case III and Case II in different channels at once (section 2.2).

---

5. What/where and the leak (question 4)

Recommendation: keep both in the main paper only as one paragraph closing Section 7, stated as a single principle with the text world, and move the detail to App. K.

- **The principle (THEORY, light).** Retrieval by place needs every token that should be retrieved from a place to sit at the same phase offset from that place, and every token that should not be retrieved there to sit elsewhere. So a token's offset must depend on its role, not its identity. The leak violates the first half (an object's step depends on its identity, so objects at one place spread over identity-dependent offsets: LEAK, registered); DirOnly violates the second (an aside's noun sits at the place's phase); learned steps satisfy both in the text world (asides offset 0.04-0.16 of a move, observed objects at a fixed offset).
- **The leak, as registered** (`LEAK_RESULTS.md`): +0.0107 in distribution (p 0.0002) for removing it, equal to MapWM's own leak, and the remedies converge where MapWM is still descending. Must be said with its replication: on fresh seeds (GAIN_PHASE D1r) SOLVED replicates (8/8 vs 0/8) but accuracy does not (+0.0045, p 0.46). Effect size ~1%, one layer, partly convergence speed (r(loss, acc) -0.94). That is a supporting result, not a contribution line.
- **What/where checks** (post hoc, `docs/WHAT_WHERE_CHECKS.md`): already in the draft's Discussion with the right status; keep it there and add that separation was shown in embeddings by MapFormer (Fig. 9), so our addition is that the trained models rely on it.
- Do not include: GAIN_PHASE's gain score (no difference, D2), NormStep on words (unmeasured), the state-change C as anything beyond location + recency.

---

6. Proposed paper text

6.1 Section 7, replacement (about 0.9 page plus figure)

> **7. Maps from words**
>
> The text world asks whether the path model can find its actions when nobody marks them. Each move of the torus walk is written as a sentence: a direction word chosen from three synonyms, verbs that do not move the walker, optional adverbs and fillers, and, after some moves, an aside that names an object without visiting it ("she thought about a cat ."). Nothing in the input says which words move the walker (vocabulary and step tables: App. J). The walks are about 132

moves long, so the map-free retrace rule of Section 4 already scores 0.934 here; we report it with the constant (0.505) and reversal-copy (0.597) floors.

One path layer learns navigation told in English. On fresh seeds (n = 6) one path layer scores 0.979 +/- 0.048 (6/6 SOLVED) against 0.505 for one-layer index RoPE (0/6), +0.474 (permutation and Fisher p 0.0022); two index layers reach 0.793 (0/6). Both index models stay below retrace on every seed; path clears it (0.993 on all eight seeds with the attention-dropout re-score, App. C). On the revisits retrace gets wrong -- places reached by a route other than undoing the last run, 5.9% of targets, best constant 0.112 -- path scores 0.961 on the fresh seeds (0.944-1.000 on every seed at ceiling) and two index layers 0.10-0.17 (analysis).

What the step table holds. On every seed, opposite direction words cancel once the component common to the four directions is removed (0.002-0.012), synonyms collapse to one step, and zeroing the direction steps drops accuracy to the constant floor (declared secondaries). The rest of the vocabulary is organised by how often a word occurs per move (analysis, 32 models, Fig. 7d): adverbs and fillers, whose count varies, get steps of 0.004-0.03 of a move; words that occur once per move (verb, seeing words, object, full stop) may carry a common tick; aside sentences get sizeable word steps that cancel within the sentence, leaving the aside's noun 0.04-0.16 of a move off the place.

The offset is why learned steps beat being told. A model whose steps are restricted to the twelve direction words stops at 0.970-0.974 on every seed, and its errors name objects told in an aside at that place: with no other word stepping, the aside's noun sits at the place's phase (Sec. 6 of `TW_NORMSTEP`). Moving each aside back onto its place's phase at evaluation, with the phase after it unchanged, drops the learned-step models from 0.989-1.000 to 0.952-0.975, with 71-100% of their errors now of the same kind; offsetting the oracle's aside words by half a move lifts it to 0.999-1.000 on every seed (analysis). "Only actions move" is the wrong separation in language: some words that are not actions need steps, so that what is mentioned is not filed where it is not.

Half the models also count moves, in channels they do not use. On 10 of 24 learned-step models a tick per move rides on every move, parallel to the verbs' step. It counts moves -- an aside sentence adds at most 0.06 of a tick, a filler word at most 0.007 -- so in Howard et al.'s terms these channels integrate distance travelled (Case II) beside the others' position (Case III). The tick alone predicts which channels drift between visits (agreement 0.98-1.00), and those channels are idle: 79-99% of retrieval attention uses channels whose tick is zero modulo pi (on the even torus a return always takes an even number of moves), and removing the 26-38 clock channels from the score costs at most 0.009 on every clock model at ceiling (7), while removing as many map channels collapses accuracy (analysis).

What must and must not move. The same rule covers the what-to-where leak on a task with new objects at every episode: a step that depends on an object's identity spreads objects at one place over identity-dependent offsets, and removing it gains +0.011 in distribution (registered, p 0.0002; on fresh seeds the

> convergence gain replicated and the accuracy gain did not, App. K). A token's offset from its place should depend on its role, not on its identity.

Figure 7: keep panels (a)-(c); add (d) the step table by word class (bars: direction differential, once-per-move words, optional words, aside words, aside net) for map and clock seeds, and (e) the aside collapse/lift test (accuracy and aside-error share). Caption must say "eval mode" for (c) as now and "re-scored" for (e).

6.2 Background 2.1, replacement of the Howard sentences

> **2.1 Rotary position as an integrator.** [first two sentences unchanged] Howard et al. (2014) drive one bank of integrators by a rate alpha(t): a constant rate codes elapsed time (Case I), the speed codes distance travelled (Case II), and the signed velocity codes position, the same for every visit to a place "like a bead on a wire" (Case III). The rotary families are these three cases with an imaginary rate: index RoPE adds a constant angle per token; CoPE, CARoPE and non-negative content steps advance by a non-negative, content-dependent amount; MapFormer, Selective RoPE and Mamba-3's data-dependent RoPE add a signed one, and alpha(t) is the selective-SSM step Delta_t. Velocity-modulated integrators have been trained in networks with a real Laplace variable (Alipour et al. 2025), and RoPE has been derived as a velocity-controlled oscillator on given coordinates (GridPE, Li et al. 2024); the machine-learning models of the Howard and Tiganj labs keep the Laplace variable real and set the complex (oscillatory) case aside as unstable (Mochizuki-Freeman et al. 2023; Kabir et al. 2025). In the rotary slot a signed rate is a bounded rotation, and the three cases differ not in what a channel can represent (Sec. 2.3) but in what the data and the optimiser make of it. TEM and TEM-t (Whittington et al. 2020, 2022) frame the Case III code as a "where" learned apart from the "what".

6.3 Discussion, replacement for "The integrator view" and "Text"

> **The integrator view.** A learned, signed, content-dependent rate can realise all three of Howard et al.'s cases, and which one a channel realises is set by what the readout keys on. Split each step into the frequency-weighted mean (a clock) and a content part. For targets keyed to a place reached at variable lags, a channel is usable only if its tick is zero modulo the lattice's return period, and the models we train put their used channels there (torus: clock part / content part 0.06-0.08; text world 0.06-0.23). For targets keyed to a count back, the used channels are clocks (k-back recency, the same one-layer architecture: 22). One text-world model can do both at once, with distance travelled in idle channels and position in used ones. In the rotary slot the cases coincide in what they can represent (Proposition 1), so the sign result of Section 6.2 is about learnability: a non-negative step could hold a Case III code and does not find one.
>
> **Text.** On ordinary text at small scale (analysis, n = 3), the channels attention uses are clock-dominated (clock part / content part 9.9-12.3 on TinyStories, 4.5-6.6 on code, against 0.06-0.23 in the text world), MapWM matches index RoPE on TinyStories (1.5053 against 1.5050 nats per token), and at matched length path integration costs on code (+0.0056 bits per byte, 3/3 seeds, t-test p 0.024). The phase grows linearly with the token index (R^2 0.994-0.999), but that alone does

> not exclude a map: text-world models that carry a perfect map show the same growth, carried by channels attention does not use. Where a map would pay in natural language, and whether a step that reads context would find it, is open.

6.4 Limitations, add

> - The text world is scripted (58 words, one movement clause per move, context-free direction words) and its walks are short enough that retrace scores 0.934; the map evidence there is the mechanism and the revisits retrace gets wrong.

6.5 Corrections to the current draft (Sections 7-8)

- Section 7 reports only the constant and reversal-copy floors; add retrace (0.934).
- "Half the seeds add a per-step clock alongside the map, in 31-38 of 64 channels, and solve as well": add that the clock counts moves (not words) and sits in channels retrieval does not use (section 2.2-2.3), else a reader takes it as a failure mode.
- "zeroing the non-direction words' steps costs it more (0.33-0.69 against 0.05-0.30)": the reason is the direction words' common component in the used channels, which the other words cancel (Spearman +0.905, section 2.2), not that the clock is used.
- Discussion "Text": "grows linearly with token index ... consistent with there being no map to build" -- replace by 6.3 (the linear growth is not diagnostic).
- References to add: Alipour et al. 2025; GridPE (Li et al., arXiv:2406.07049); Mochizuki-Freeman et al. 2023; Kabir et al. 2025 (arXiv:2412.15292); optionally Sarkar and Howard 2026 (arXiv:2601.02618), RoMem (arXiv:2604.11544).

---

7. Proposed GPU experiments (with cost)

Costs from TW_STATECHANGE (32 text-world runs, 900 epochs, T = 1024, 1 layer, on one GPU in 5.7 h at 2 jobs/GPU, about 0.18 GPU-h per run) and TW_AMBIG's own estimate (64 runs at 1800 epochs, ~15 h). Each needs the rule-29 audit and a pre-registration before launch; pilots on outside seeds.

id	question	design	cost	priority
E1	Why do half the seeds build an idle move counter? Hypothesis: it appears early (when recency -- reversal copying, floor 0.597 -- is the cheapest cue) and attention later leaves its channels rather than unlearning it	8 fresh MapWM text-world seeds, 900 epochs, step table and per-channel usage saved every 25 epochs (no change to training). Registered readouts: epoch at which the tick exceeds 1 move and the map's opposition falls below 0.05; usage of later-clock channels over time	8 runs, ~1.5 GPU-h	medium (explains a figure panel; not a claim)
E2	Does a context-dependent step learn which uses of a word move (TW_AMBIG,	direction words also used as non-moving content; MapWM vs	64 runs at 1800 epochs, ~15 h	low for this paper (a separate question;

id	question	design	cost	priority
	prepared, STOPPED 2026-10-11)	oracle vs HSR (hidden-state step), cue near/far	(prereg's estimate)	mechanisms published, LIT_CONTEXT_STEPS)
E3	Does the used-channel ratio move with the readout inside one stream? Double dissociation	text world where a fraction of targets ask "what did she see k moves ago" (count-keyed) and the rest are revisits (place-keyed); 8 seeds. Prediction: used channels split into rho << 1 and rho >> 1, and ablating each set removes only its query type	16 runs (with a revisit-only control), ~3 GPU-h	high if the theory paragraph is kept: it turns section 3.3's across-task comparison into a within-model test
E4	Is the text-world advantage over retrace a map at long walks?	text world at 4096 words (~530 moves), where retrace should fall as on the torus (0.843 at 1024 moves); path vs RoPE 1-2 layers, 8 seeds	24 runs at 4x tokens per run (attention cost grows faster than 4x), ~20-35 GPU-h; --fast-attn valid (no MapEM)	medium: removes the short-walk caveat of 2.6
E5	Retrace floor and strata re-scored on TW_NORMSTEP / TW_STATECHANGE batches	CPU, no training	0 GPU, ~1 h CPU	do before any text-world number goes in the paper

---

8. Analyses run (scripts and outputs)

All in `docs/paper/expansion/scripts/language/`; run from `/home/prashr` with `PYTHONPATH=/home/prashr python3